Regression with a categorical predictor

Reminders and agenda

- Exam 1 next Friday (Feb 27)
  - Covers material through the last class (Feb 16)
- Exam 1 review day next Wednesday (Feb 25)
- Studying advice:
  - Practice questions on the course website
  - Review previous class activities
  - Review previous homework assignments
- Today: thinking about categorical explanatory variables

Regression so far

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

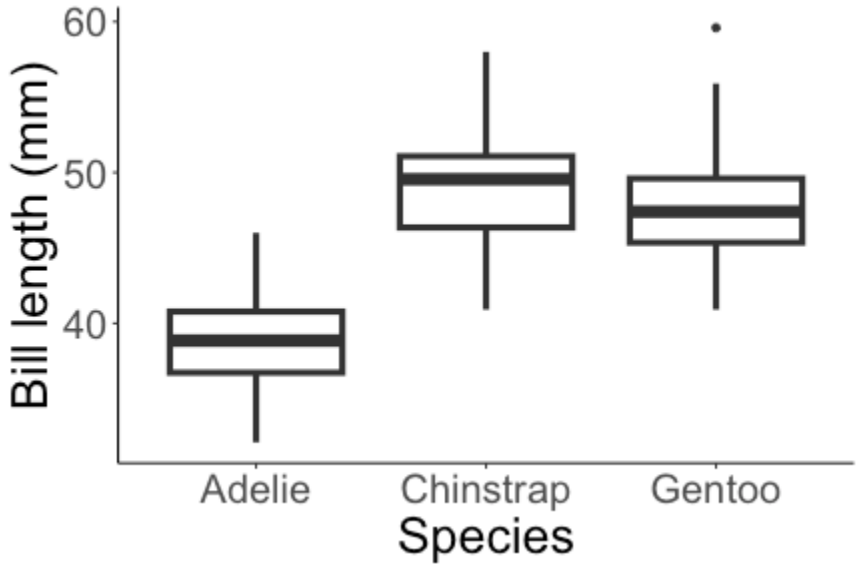
Polynomial regression:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \varepsilon$$

where x and y are both **quantitative**.

> What if x is *categorical*?

Categorical predictor



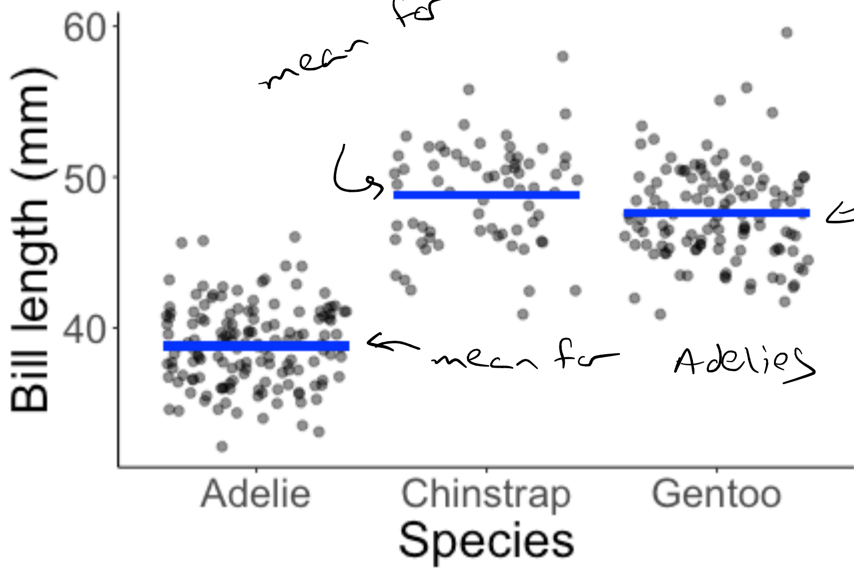
- How do we model the relationship between species and bill length?
- Can we test for a relationship between bill length and species?

wrong: $y = \beta_0 + \beta_1 \text{ Species } + \varepsilon$

what does a "one unit" increase in Species mean?

Categorical predictor

Imagine population like this:



mean for Chinstraps

mean for Gentoos

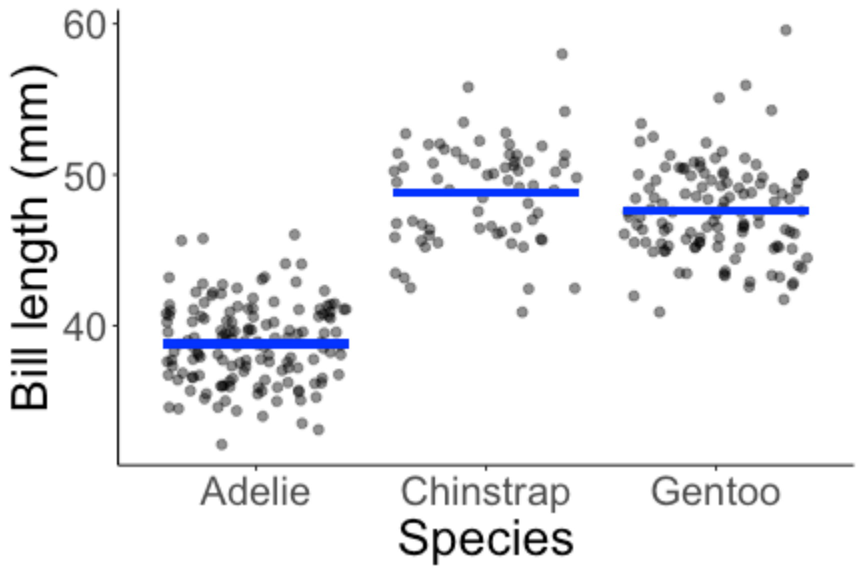
mean for Adelies

If species $=$ Adelie,

$$\text{bill length} = \text{mean Adelie bill length} + \varepsilon$$

Categorical predictor

Imagine population like this:

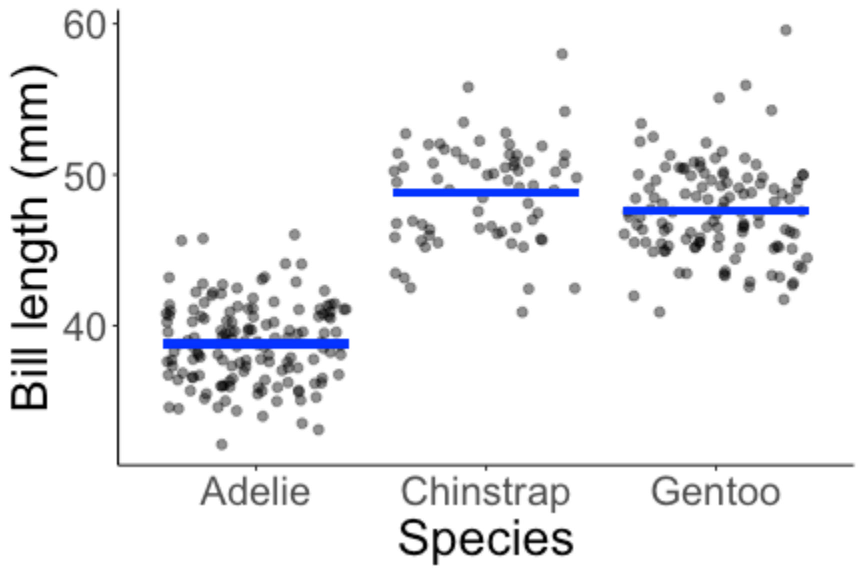


If species $= \text{Chinstrap}$,

$$\text{bill length} = \text{mean Chinstrap bill length} + \varepsilon$$

Categorical predictor

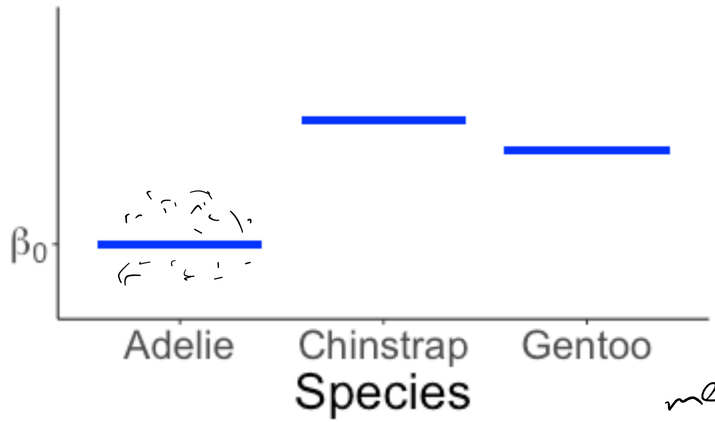
Imagine population like this:



If $\text{species} = \text{Gentoo}$,

$$\text{bill length} = \text{mean Gentoo bill length} + \varepsilon$$

Categorical predictor



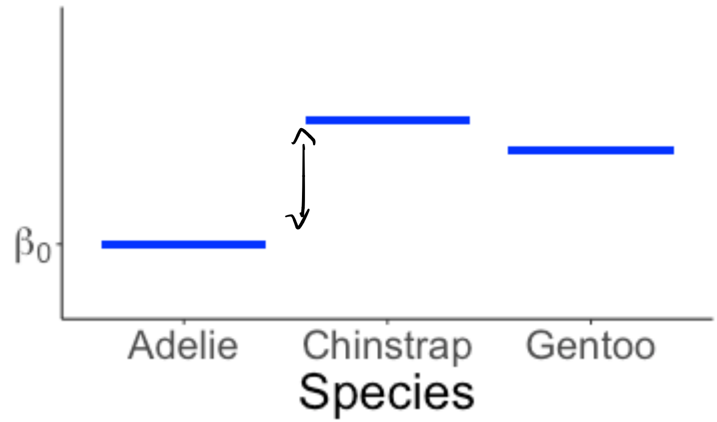
mean for Adelies

variability in population

If species $=$ Adelie, $\quad \text{bill length} = \beta_0 + \varepsilon$

- β_0 = true mean bill length of Adelie penguins (in the population)

Categorical predictor

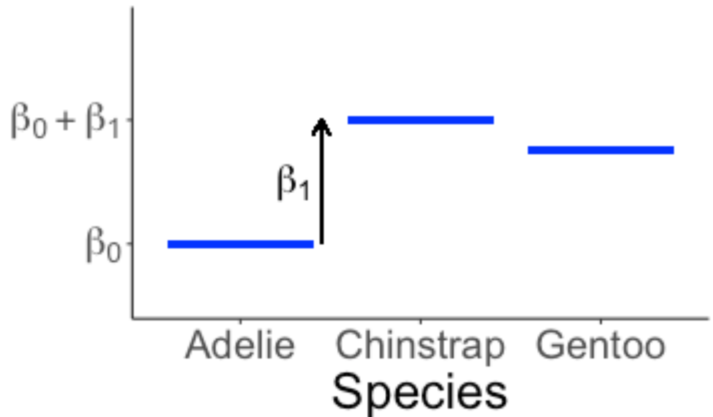


$$\text{If species} = \text{Adelie}, \qquad \text{bill length} = \beta_0 + \varepsilon$$

- β_0 = true mean bill length of Adelie penguins (in the population)

> What notation could I use for Chinstrap penguins?

Categorical predictor



for Adelies

$$\text{If species} = \text{Adelie}, \quad \text{bill length} = \underbrace{\beta_0}_{\text{mean}} + \varepsilon$$

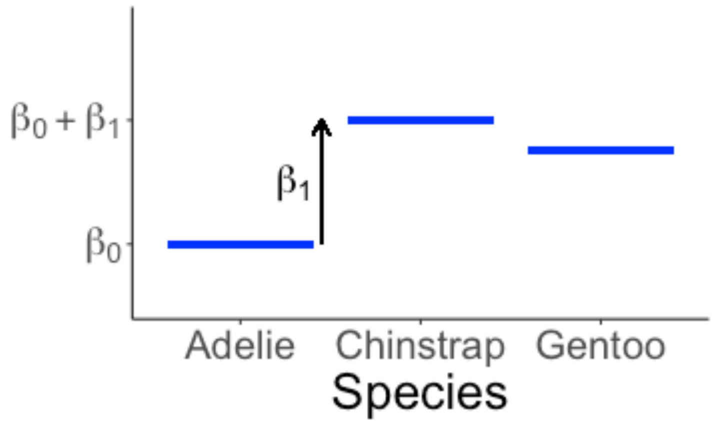
- β_0 = true mean bill length of Adelie penguins (in the population)

for chinstrap penguins

$$\text{If species} = \text{Chinstrap}, \quad \text{bill length} = \overbrace{\beta_0 + \beta_1}^{\text{mean}} + \varepsilon$$

β_1 = difference in mean bill length between Adelie and Chinstrap penguins

Categorical predictor

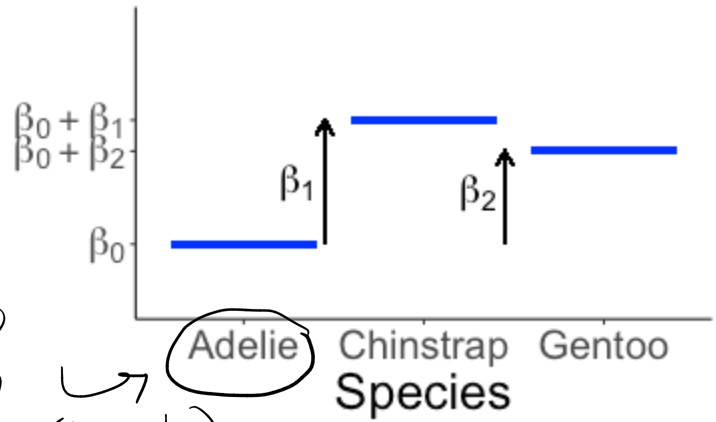


If species $= \text{Adelie}, \quad \text{bill length} = \beta_0 + \varepsilon$

If species $= \text{Chinstrap}, \ \text{bill length} = \beta_0 + \beta_1 + \varepsilon$

> What about Gentoo penguins?

Categorical predictor



reference group
(aka reference level,
baseline level,
baseline category, etc.)

$$\text{If species} = \text{Adelie}, \quad \text{bill length} = \beta_0 + \varepsilon$$

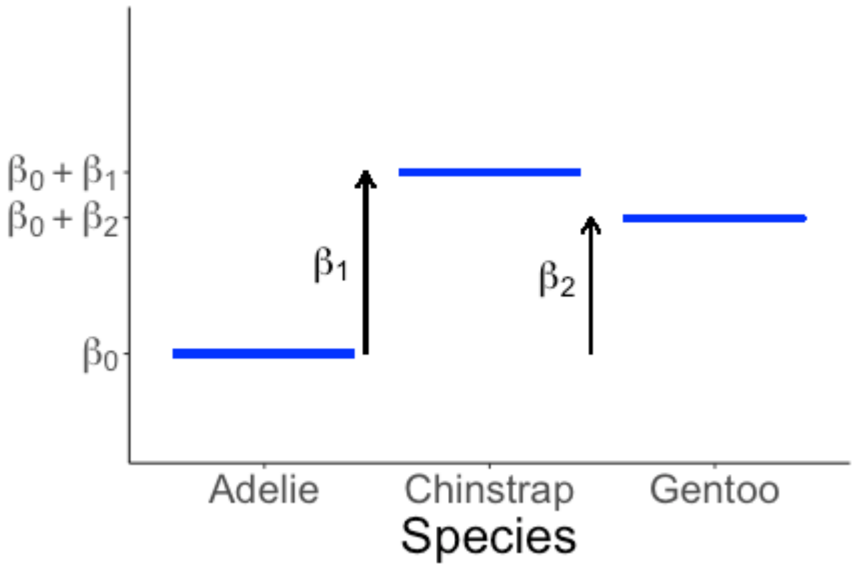
$$\text{If species} = \text{Chinstrap}, \quad \text{bill length} = \beta_0 + \beta_1 + \varepsilon$$

$$\text{If species} = \text{Gentoo}, \quad \text{bill length} = \beta_0 + \beta_2 + \varepsilon$$

β_2 = mean difference in bill length between Gentoo & Adelie penguins

In this model: all coefficients are relative to reference group (Adelie penguins)

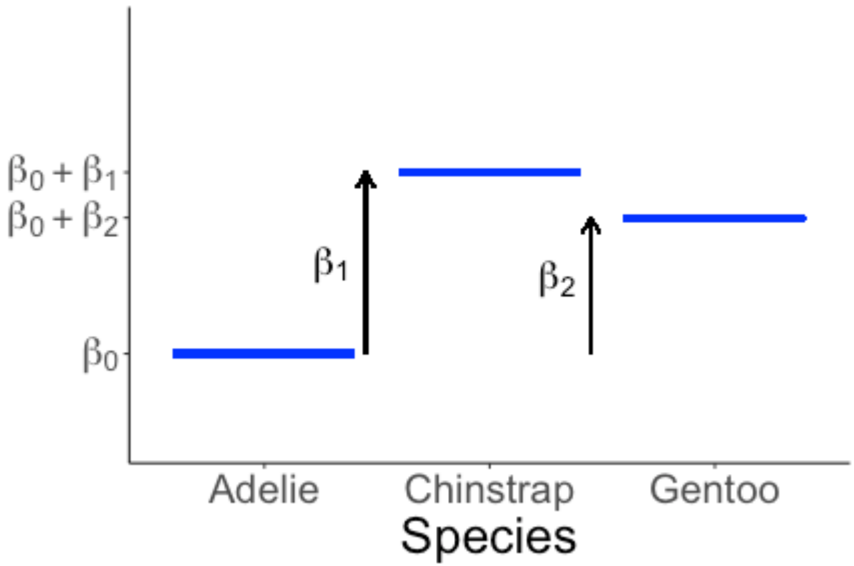
Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

(piecewise)

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

> Can we write this more concisely?

Indicator variables

Take values 0 or 1, depending on a logical statement

$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

> Do I need another indicator variable to represent Adelie penguins?

No: Adelie is what we get when IsChinstrap = 0 and IsGentoo = 0

In general: represent a categorical variable that has K levels with K-1 indicator variables

Indicator variables

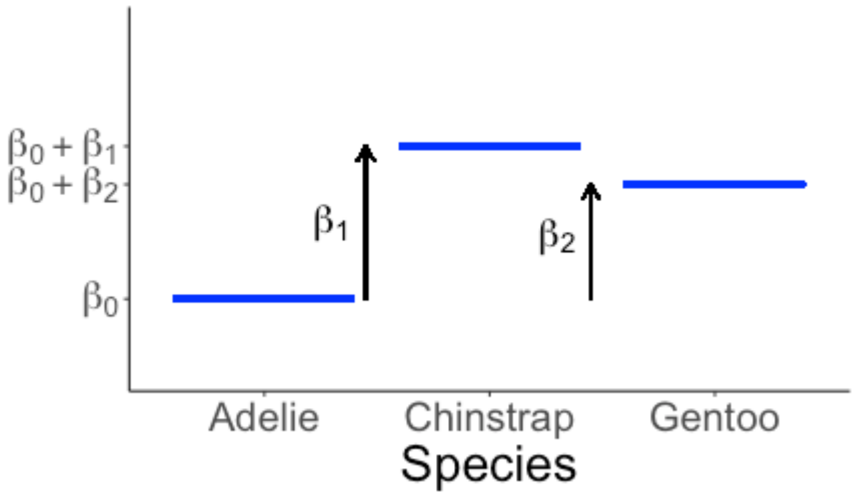
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

Species	IsChinstrap	IsGentoo
Adelie	0	0
Chinstrap	1	0
Gentoo	0	1

Indicator variables



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01\ \text{IsChinstrap} + 8.74\ \text{IsGentoo}$$

- What is the estimated average bill length for each of the three species of penguin?
- There are 333 penguins in the dataset used to fit the model. How many degrees of freedom does the fitted model have?

Adelie : $\widehat{\text{bill length}} = 38.82 + 10.01(0) + 8.74(0)$

$= 38.82$

Chinstrap : $\widehat{\text{bill length}} = 38.82 + 10.01(1) + 8.74(0)$

$= 38.82 + 10.01 = 48.83$

Gentoo : $\widehat{\text{bill length}} = 38.82 + 8.74 = 47.56$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01\ \text{IsChinstrap} + 8.74\ \text{IsGentoo}$$

Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	38.82 + 10.01 = 48.83
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Estimated model

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Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	38.82 + 10.01 = 48.83
Gentoo	38.82 + 8.74 = 47.56

Degrees of freedom: 333 - 3 = 330

- (Lose one degree of freedom for each parameter estimated)

$\beta_0, \beta_1, \beta_2$

Fitting the model in R

R will automatically create indicator variables:

```
lm(bill_length_mm ~ species, data = penguins_no_nas)
```

response

explanatory (categorical)

```
##
## Call:
## lm(formula = bill_length_mm ~ species, data = penguins_n
##
## Coefficients:
##       (Intercept)  speciesChinstrap     speciesGentoo
##            38.824            10.010             8.744
```

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Class activity

- Practice regression with a single categorical predictor, with experimental data on fruit flies

https://sta112-s26.github.io/class_activities/ca_15.html